

INFUSION v5.1 Change Request Validation and Approval Pack

Prepared for Development Team approval - no main application files changed in this review document.

Reviewed source: INFUSION_CHANGE_REQUESTS_v5.1.pdf, 14 pages. **Current dashboard baseline observed:** v5.1.0-swing-horizon-map. **Review date:** 2026-08-02 IST.

Executive decision: The change requests are broadly valid and should be adapted, but shipped as a disciplined v5.2 "Reality Check" release before adding more UI/AI features. The most urgent corrections are option cost/slippage, delta-derived premium SL, breakeven-vs-ATR gating, honest auto-journal logging, and analytics reset.

Area	Validation result	Approval recommendation
P0 blockers	Valid. Current code still has a hardcoded 28% option SL and lacks a central cost model, target-breakeven gate, 60-session IV-rank gate, and strict delta band.	Approve first. These determine whether option trades are realistically tradable.
P1 data integrity	Valid. Journal is manual/selective and analytics still headline precision/optimizer outputs.	Approve second. Without honest auto-logging, backtest/analytics remains biased.
P2 safety	Mostly valid. Physical settlement and liquidity whitelist are important for Indian stock options. News keyword logic should be demoted/replaced with event calendar.	Approve after P0/P1. Some event data sources may need manual CSV/API support.
P3 signal cleanup	Valid but requires data study before changing scores. Regime gating exists partially but is not as strict as requested.	Approve as research + controlled deployment. Do not blind-change conviction formula.

Critical caution: These changes improve measurement truth and execution realism. They do not prove strategy edge or guarantee win rate. The goal is to stop false confidence, not to force more trades.

1. Current-Code Evidence Summary

Finding	Current evidence	Impact
Hardcoded option stop	services/api/src/api/routes/execution.py uses option_sl_pct = 28.0 and estimated_option_sl = premium * (1 - option_sl_pct / 100).	Invalid for stock options because premium risk depends on underlying SL distance, strike delta, spread, and expiry.
No central cost model	Journal, execution, and analytics do not call a shared cost_model. Journal stores premium/spread but not gross_pnl, total_costs, or net_pnl.	P&L; and expectancy can be optimistic because charges and bid/ask crossing are missing.
Manual journal logging	Dashboard scanner calls POST /api/journal/trades only from Log Paper Trade. Journal source is dashboard_manual_paper.	Dataset is selection-biased; skipped signals are not measured.
Precision still headline metric	analytics.js renders Precision, Phase 5 Proof, 80% Precision Optimizer, and Forward Proof. analytics.py exposes /api/analytics/precision endpoints.	Can mislead decision making; expectancy and cost drag should be primary.
Optimizer still active in UI	analytics.js subscribes to /api/backtest/optimize?days=60■=80&min;_decided=50.	Potential curve-fit risk if used to influence live thresholds.
Option chain scoring partial	market.py scores OI, IV, spread, strike and a loose delta-like band 0.30-0.70, but no 60-session IV rank, breakeven-vs-ATR gate, or cost-per-unit target gate.	Can mark contracts attractive even when expected move cannot pay premium + spread + costs.
News edge keyword logic	news.py contains POSITIVE_WORDS, NEGATIVE_WORDS, EVENT_RISK_WORDS and computes headline score.	Useful context only, but not robust enough as trading edge. Event calendar is safer.
Regime gate partial only	suppression.py blocks volatile regime, but does not hard block CE in risk-off, PE in risk-on, or all entries in neutral/chop.	Signals can pass against broader market conditions.

2. Change Request Validity Matrix

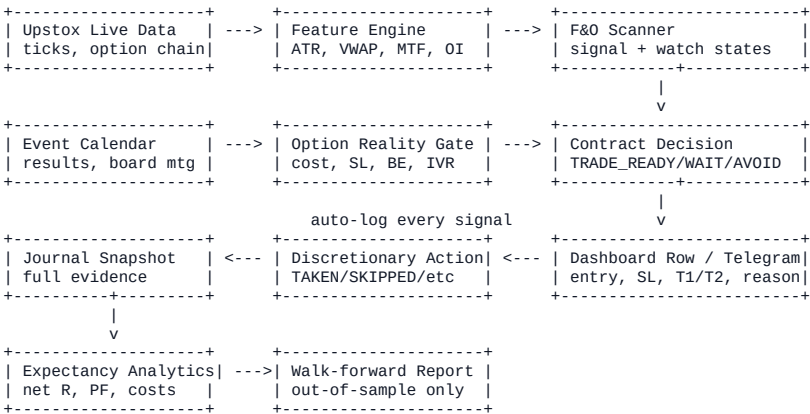
ID	Request	Validity	Decision	Impact / risk
CR-1	Central cost & slippage model	Valid	Adopt P0	Creates one truth for P&L.; Requires execution, journal, analytics, Telegram recap changes.
CR-2	Remove 28% premium SL	Valid - confirmed	Adopt P0	Fixes major risk sizing error. Needs delta from selected option and guards.
CR-3	Breakeven vs expected-move gate	Very high impact	Adopt P0	Blocks contracts where T1 cannot pay premium/spread/cost. May reduce trade count but improves realism.
CR-4	IV rank gate	Valid but data-dependent	Adopt P0/P1 bridge	Needs 60-session IV history store. Until history exists, chain should be CHAIN_PENDING, not guessed.
CR-5	Delta band 0.35-0.60	Valid	Adopt P0	Current 0.30-0.70 is looser. Stricter band suits directional option buying.
CR-6	Auto-log every signal	Valid	Adopt P1	Creates honest dataset. Log Paper Trade becomes action marker, not creator.
CR-7	Reset analytics baseline	Valid	Adopt P1	Legacy metrics become separated. Avoid mixing contaminated and post-fix data.
CR-8	Replace precision with expectancy	Valid	Adopt P1	Better decision metrics. Precision remains secondary.
CR-9	Disable/rebuild optimizer	Valid	Adopt P1	Stops curve-fit thresholds. Rebuild only as walk-forward out-of-sample report.
CR-10	Physical settlement rule	Valid	Adopt P2	Indian stock options need expiry-week safety. Requires positions snapshot if/when live orders exist.
CR-11	Liquidity whitelist	Valid	Adopt P2	Requires rolling option OI/volume/spread history. Prevents illiquid contract traps.
CR-12	Event calendar replaces news keywords	Valid direction	Adopt P2	Safer than sentiment. Needs NSE/corporate event source or manual upload workflow.
CR-13	De-correlate conviction score	Valid research item	Adopt P3 report first	Do not alter scores before correlation matrix over stored scanner features.
CR-14	Market regime as hard gate	Valid but calibrate	Adopt P3 / maybe P1 soft gate	Hard neutral block may starve trades. Use paper mode first, with override label.
CR-15	Feature freeze	Valid	Adopt immediately	Freeze new features until P0-P1 shipped and paper data is clean.

3. Recommended Release Plan

Release	Scope	Exit criteria
v5.2.0 Reality Check - P0	Cost model, delta-derived SL, breakeven gate, delta band, IV-rank storage design, option basis UI fields.	0% premium move produces negative net P&L; no 28% constant; far OTM low ATR contract blocked; staged ticket shows costs and premium-risk fields.
v5.2.1 Honest Journal - P1	Auto-log every signal, discretionary action states, legacy analytics isolation, expectancy metrics, optimizer disabled.	Every fired signal has journal row. Analytics separates TAKEN/SKIPPED/NOT_REVIEWED. Precision demoted.
v5.3.0 Option Safety - P2	Physical settlement guard, liquidity whitelist, event calendar replacing keyword news edge.	Expiry-week ITM risk banner works. Illiquid instruments visible but AVOID_CONTRACT. Event date visible on stock row.
v5.4.0 Score Science - P3	Correlation matrix report, de-correlated conviction formula, regime hard/soft gate experiment.	Published one-off factor correlation report. New score formula approved after paper validation.

Recommended approval wording: Approve P0 and P1 immediately. Approve P2 and P3 as backlog after P0/P1 acceptance. Freeze all non-essential feature additions until v5.2.1 is live and stable.

4. Target Data Flow and UI Model Diagram



Dashboard region	Proposed v5.2 display change
Top market strip	Regime, breadth, NIFTY/BANKNIFTY/STOCK LTP, token state, physical-settlement red banner if expiry-week risk exists.
F&O; scanner row	Keep fixed stock/sector columns. Add Req Move %, Exp Move %, BE price, Delta, IV Rank, Cost %, Contract Status. Selected row expands inline.
Selected stock AI trade map	Show: Breakout above, 5M/15M sustain rule, move-to T1/T2, breakdown below, option breakeven, premium SL, why blocked/ready.
Option Basis strip	Ask entry, bid exit assumption, spread, delta, IV rank, OI/volume whitelist, premium risk %, required move vs ATR.
Journal tab	Auto logged signals with discretionary_action. User changes NOT_REVIEWED to TAKEN or SKIPPED with reason.
Analytics tab	Headline Expectancy, Profit Factor, Cost Drag, Max Drawdown, R distribution. Precision becomes secondary.

5. Proposed Code Contracts - P0

These snippets are approval contracts, not applied patches. They show the implementation direction the development team would review before touching main files.

5.1 New module: `services/api/src/api/cost_model.py`

```
from dataclasses import dataclass

@dataclass(frozen=True)
class OptionCostAssumptions:
    brokerage_per_order: float = 20.0
    stt_sell_rate: float = 0.000625
    exchange_txn_rate: float = 0.0003503
    sebi_rate: float = 0.000001
    gst_rate: float = 0.18
    stamp_buy_rate: float = 0.00003

@dataclass(frozen=True)
class OptionTradeCostInput:
    side: str
    entry_ask: float
    exit_bid: float
    bid_at_entry: float
    ask_at_exit: float
    quantity: int
    lot_size: int
    assumptions: OptionCostAssumptions = OptionCostAssumptions()

def compute(trade: OptionTradeCostInput) -> dict:
    buy_turnover = trade.entry_ask * trade.quantity
    sell_turnover = trade.exit_bid * trade.quantity
    gross_pnl = (trade.exit_bid - trade.entry_ask) * trade.quantity
    brokerage = trade.assumptions.brokerage_per_order * 2
    stt = sell_turnover * trade.assumptions.stt_sell_rate
    exchange = (buy_turnover + sell_turnover) * trade.assumptions.exchange_txn_rate
    sebi = (buy_turnover + sell_turnover) * trade.assumptions.sebi_rate
    gst = (brokerage + exchange) * trade.assumptions.gst_rate
    stamp = buy_turnover * trade.assumptions.stamp_buy_rate
    slippage_cost = max(0.0, trade.entry_ask - trade.bid_at_entry) * trade.quantity
    total_costs = brokerage + stt + exchange + sebi + gst + stamp + slippage_cost
    premium_paid = max(buy_turnover, 1.0)
    return {
        'gross_pnl': round(gross_pnl, 2),
        'brokerage': round(brokerage, 2),
        'stt': round(stt, 2),
        'exchange_charges': round(exchange, 2),
        'sebi_fees': round(sebi, 2),
        'gst': round(gst, 2),
        'stamp_duty': round(stamp, 2),
        'slippage_cost': round(slippage_cost, 2),
        'total_costs': round(total_costs, 2),
        'net_pnl': round(gross_pnl - total_costs, 2),
    }
```

```
    'cost_as_pct_of_premium': round(total_costs / premium_paid * 100, 3),  
}
```

5.2 New module: services/api/src/api/option_reality.py

```
def derive_option_sl(entry_underlying: float, sl_underlying: float,
                    entry_premium_ask: float, delta: float) -> dict:
    underlying_risk = abs(entry_underlying - sl_underlying)
    premium_risk = underlying_risk * abs(delta)
    premium_risk_pct = premium_risk / max(entry_premium_ask, 0.01)
    option_sl_price = max(0.05, entry_premium_ask - premium_risk)
    blockers = []
    if premium_risk_pct < 0.12:
        blockers.append('SL inside spread - noise stop')
    if premium_risk_pct > 0.45:
        blockers.append('strike too far OTM for this SL distance')
    return {
        'underlying_risk': round(underlying_risk, 2),
        'delta_used': round(delta, 4),
        'premium_risk': round(premium_risk, 2),
        'premium_risk_pct': round(premium_risk_pct * 100, 2),
        'option_sl_price': round(option_sl_price, 2),
        'hard_blockers': blockers,
    }

def breakeven_gate(bias: str, spot: float, strike: float, entry_ask: float,
                  spread_per_unit: float, est_costs_per_unit: float,
                  daily_atr_pct: float, expected_holding_days: float,
                  underlying_t1: float) -> dict:
    if bias == 'CE':
        breakeven = strike + entry_ask + spread_per_unit + est_costs_per_unit
    else:
        breakeven = strike - entry_ask - spread_per_unit - est_costs_per_unit
    required_move_pct = abs(breakeven - spot) / max(spot, 0.01) * 100
    expected_move_pct = daily_atr_pct * expected_holding_days
    blockers, hard = [], []
    if expected_move_pct <= 0:
        hard.append('ATR expected move unavailable')
    elif required_move_pct > 1.0 * expected_move_pct:
        hard.append('breakeven too far vs ATR')
    elif required_move_pct > 0.6 * expected_move_pct:
        blockers.append('breakeven too far vs ATR')
    if (bias == 'CE' and underlying_t1 <= breakeven) or (bias == 'PE' and underlying_t1 >= breakeven):
        hard.append('target does not pay for premium')
    return {
        'breakeven_underlying': round(breakeven, 2),
        'required_move_pct': round(required_move_pct, 3),
        'expected_move_pct': round(expected_move_pct, 3),
        'blockers': blockers,
        'hard_blockers': hard,
    }

def delta_band_gate(delta: float) -> dict:
    ok = 0.35 <= abs(delta) <= 0.60
    return {'delta_band_pass': ok, 'reason': '' if ok else 'delta outside directional band'}
```

6. Proposed API Contracts - P1/P2/P3

6.1 Auto journal API shape

```
POST /api/journal/signals/auto
{
  'signal_id': '20260802_101533_NAUKRI_options_first_hybrid',
  'source': 'scanner_auto_signal',
  'discretionary_action': 'NOT_REVIEWED',
  'symbol': 'NAUKRI',
  'decision': 'BUY CE',
  'entry': 1003.4,
  'stop': 1001.6,
  'target1': 1008.5,
  'target2': 1011.2,
  'selected_option': {...},
  'reality_snapshot': {
    'delta': 0.48,
    'iv_rank': 42.0,
    'premium_risk_pct': 23.2,
    'underlying_risk': 1.8,
    'required_move_pct': 0.38,
    'expected_move_pct': 0.74,
    'breakeven_underlying': 1006.1,
    'spread_at_decision': 0.9,
    'est_costs_per_unit': 0.15,
    'gross_pnl': 0,
    'total_costs': 0,
    'net_pnl': 0,
    'regime_at_decision': 'RISK_ON',
    'next_event_date': '2026-08-07',
    'liquidity_whitelist_pass': True
  }
}

PATCH /api/journal/trades/{id}/discretion
{
  'discretionary_action': 'TAKEN' | 'SKIPPED',
  'skip_reason': 'Spread widened after alert'
}
```

6.2 Analytics replacement endpoints

```
GET /api/analytics/expectancy?baseline=post_reality
{
  'expectancy_r': 0.42,
  'profit_factor': 1.84,
  'avg_win_r': 1.9,
  'avg_loss_r': -0.74,
  'cost_drag_pct': 18.6,
  'max_drawdown_r': -4.2,
  'hit_rate': 0.48,
  'sample': {'total': 120, 'taken': 36, 'skipped': 52, 'not_reviewed': 32}
}

GET /api/analytics/walk_forward
{
  'display_mode': 'out_of_sample_only',
  'windows': [{'train': '2026-08,2026-09', 'validate': '2026-10', 'expectancy_r': 0.31}],
  'live_threshold_write_enabled': False
}
```

6.3 Event calendar replacement for news edge

```
GET /api/events/stock?symbol=NAUKRI
{
  'symbol': 'NAUKRI',
  'next_event_date': '2026-08-08',
  'event_type': 'RESULTS',
  'days_to_event': 2,
  'entry_allowed': False,
  'block_reason': 'hard block from T-2 to T+1 around results'
}
```

6.4 Correlation report before score changes

```
python tools/scanner_correlation_report.py --from 2026-08-01 --to 2026-10-31 \
--features vwap_state,ema_stack,macd,rsi,mtf_1m,mtf_5m,mtf_15m,mtf_1h,mtf_4h,mtf_1d,bb_compression \
--output output/pdf/infusion_factor_correlation_report.pdf

Rule: if abs(correlation) > 0.70, keep the factor with better standalone out-of-sample expectancy.
```


7. Required Journal Snapshot Fields

Field	Why needed	Source
delta	Premium SL and delta band realism	Upstox option greeks
iv_rank	Avoid buying high IV crush	Rolling 60-session IV store
premium_risk_pct	Block noisy or too-wide premium stops	option_reality.derive_option_sl
underlying_risk	Explain premium SL derivation	scanner entry/SL
required_move_pct	Move needed for breakeven	breakeven_gate
expected_move_pct	ATR-based move expectation	feature engine ATR * holding days
breakeven_underlying	Check if T1 pays premium	strike + premium + spread + costs
spread_at_decision	Slippage/liquidity attribution	option chain bid/ask
est_costs_per_unit	True option breakeven	cost_model
gross_pnl	Raw result before charges	cost_model
total_costs	Brokerage/tax/slippage drag	cost_model
net_pnl	Actual paper result	cost_model
regime_at_decision	Regime attribution	/api/regime
discretionary_action	Taken/skipped/not reviewed	journal action
skip_reason	Trader feedback loop	UI form
next_event_date	Avoid event risk traps	event calendar
liquidity_whitelist_pass	Tradable contract filter	liquidity service

8. UI Acceptance Model

```
HEADER: NIFTY | BANKNIFTY | TOKEN | REGIME | BREADTH | PHYSICAL SETTLEMENT ALERT

[ MARKET PULSE / SECTOR HEATMAP ]
all F&O sectors visible; green/yellow/red strength; sortable by strength delta

[ SELECTED STOCK REALITY CARD ]
SYMBOL LTP ACTION HORIZON
Breakout above: 1234.50 | Sustain: 15M close above + volume > 1.5x
Move to: 1250.00 / 1268.00 | Breakdown below: 1218.00
Option: 1240 CE | Ask entry: 18.50 | Derived SL: 14.30 | Delta: 0.48
Req move: 0.62% | Exp move: 1.10% | BE: 1258.65 | IV Rank: 42 | Cost: 6.8%
Verdict: TRADE_READY / WAIT_CONTRACT / AVOID_CONTRACT with exact reasons

[ F&O SCREENER TABLE ]
Stock | Sector | LTP | VWAP | RSI | Vol | MTF | Strength | Conviction | Horizon
Delta | IVR | Req Move | Exp Move | BE | Cost % | Option Status | Reason

[ TABS ]
1. Option Basis 2. Journal 3. Analytics 4. Event Calendar 5. Safety Cockpit
```

9. Final Approval Checklist

Approval item	Approve?	Notes
Proceed with v5.2.0 P0 implementation before any more UI features	☐ Yes / ☐ No	Recommended YES
Remove hardcoded 28% option SL	☐ Yes / ☐ No	Confirmed current defect
Implement cost model with ask entry and bid exit assumptions	☐ Yes / ☐ No	Required for honest P&L;
Block contracts where underlying T1 does not clear breakeven	☐ Yes / ☐ No	High impact for option buying
Auto-log every signal and change Log Paper Trade to TAKEN marker	☐ Yes / ☐ No	Required for unbiased analytics
Separate legacy contaminated analytics from post-fix analytics	☐ Yes / ☐ No	Do not delete old data
Demote precision and headline expectancy/profit factor/cost drag	☐ Yes / ☐ No	Recommended YES
Disable 80% optimizer influence until walk-forward rebuilt	☐ Yes / ☐ No	Recommended YES
Approve P2/P3 as backlog after P0/P1 acceptance	☐ Yes / ☐ No	Recommended YES

Conclusion: The development AI change requests are technically valid and should be adapted. The only adjustment is sequencing: P0 and P1 must be completed before P2/P3 and before any further feature expansion. This keeps Infusion focused on realistic option trading decisions rather than visually impressive but statistically unreliable outputs.